Forecasting Flight Delays at RDU Airport Using Time Series Models

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1 Introduction, Motivation, Relevance, and Objectives

1.1 Background and Context

Flight delays remain one of the most persistent operational challenges in commercial aviation, with substantial economic and social consequences. According to the U.S. Bureau of Transportation Statistics (BTS), arrival delays directly translate into lost productivity, missed connections, increased fuel consumption, and diminished passenger satisfaction. At the national level, the Federal Aviation Administration has estimated that the total cost of flight delays to the U.S. economy exceeds tens of billions of dollars annually, highlighting the value of reliable delay forecasting for both industry stakeholders and public policy.

Raleigh–Durham International Airport (RDU) serves the Research Triangle region of North Carolina, one of the fastest-growing metropolitan areas in the United States. RDU handles millions of passengers each year and continues to expand its route network to accommodate rising demand. As regional air traffic grows, the ability to anticipate periods of elevated delay activity becomes increasingly important for airport operations, airline scheduling, and passenger decision-making.

1.2 Motivation and Relevance

Three considerations motivated the choice of RDU as the study site. First, RDU is a mid-sized hub whose delay dynamics are shaped by a diverse mix of carriers, which provides a representative case for generalizable forecasting methods. Second, the Research Triangle's sustained population and economic growth make accurate delay forecasts operationally valuable for regional businesses, commuters, and airport authorities. Third, the BTS provides consistent, long-horizon monthly data on on-time performance at the airport–carrier level, enabling a sufficiently long training history for statistical time series modeling.

Accurate forecasts of monthly delay rates support several downstream decisions. Airlines and ground-handling operators can adjust staffing, gate allocation, and aircraft rotations in anticipation of high-delay months. Airport management can plan capacity, communication campaigns, and irregular-operations protocols around predicted seasonal peaks. Passengers and regional logistics operators can incorporate expected delay patterns into travel and shipping decisions. Because these operational decisions are often made on a monthly planning cycle, a monthly-resolution forecast aligns naturally with real-world planning horizons.

1.3 Objectives

The objectives of this project are as follows:

1. Construct a clean, monthly-resolution time series of arrival delay rates at RDU using BTS carrier-level records from January 2008 through December 2019.
2. Characterize the time series structure at RDU, including stationarity, trend, and seasonality, and rigorously evaluate whether identified seasonal structure reflects genuine signal rather than an artifact of the decomposition method.
3. Fit and compare a range of forecasting models spanning classical, state-space, regression-based, and machine-learning approaches: ETS, SARIMA, ARIMA with Fourier terms, TBATS, NNETAR, and Prophet.
4. Evaluate model performance on a held-out 2019 test set using multiple accuracy metrics (RMSE, MAE) and diagnose model residuals to assess whether each model has captured the underlying temporal structure.
5. Identify the most suitable model for operational forecasting at RDU and document the limitations of the chosen approach, including implications for post-pandemic extensions of the analysis.

The full code base, data-processing pipeline, and forecasting files supporting this report are available in the project's public GitHub repository: https://github.com/lingyuehao/Fischer_Hao_Chen_ENV797_TSA_FinalProject.git.

2 Dataset Information

2.1 Data Source

The data used in this study were obtained from the U.S. Bureau of Transportation Statistics (BTS), specifically the *Airline On-Time Performance* database maintained by the Office of Airline Information. This database provides monthly airport–carrier records of arrival performance for all U.S. domestic flights operated by certified carriers that account for at least one percent of domestic scheduled passenger revenue.

Records for Raleigh–Durham International Airport (RDU) across all reporting carriers were extracted for the period **January 2008 through December 2019**. The year 2019 was chosen as the cutoff to preserve a complete pre-pandemic annual cycle for testing, avoiding the structural break introduced by COVID-19 in 2020 that would violate the stationarity assumptions of the forecasting models considered.

2.2 Raw Data Structure

Each row of the raw dataset corresponds to a single carrier–month–airport combination. The key variables used in this analysis are summarized in Table 1.

Table 1: Description of key variables in the RDU flight delay dataset.

Variable	Type	Units	Description
year	Integer	Calendar year	Year of observation (2008-2019)
month	Integer	Month (1-12)	Month of observation
carrier	Character	IATA code	Two-letter airline identifier
carrier_name	Character	Full name	Airline name corresponding to carrier code
airport	Character	IATA code (RDU)	Airport identifier (filtered to RDU only)
arr_flights	Numeric	Flights	Number of arriving flights at RDU by carrier
arr_del15	Numeric	Flights	Arrivals delayed by 15+ minutes
delay_rate	Numeric (derived)	Proportion (0-1)	sum(arr_del15) / sum(arr_flights)

The raw training-period file (2008–2018) and the test-period file (2019) both contained carrier-level monthly records across 21 variables. No missing values were detected in either file across any variable, eliminating the need for imputation.

2.3 Data Wrangling and Time Series Construction

The raw data were processed into a single-series, monthly-resolution time series through the following pipeline:

1. **Filter to valid records.** Carrier–month records with `arr_flights = 0` were removed, as these correspond to months in which a carrier did not operate at RDU and contain no information about delay incidence.

2. **Aggregate across carriers.** For each `(year, month)` pair, the total number of arriving flights and the total number of delayed arrivals (15 minutes or more) were summed across all reporting carriers:

$$\text{arr_flights}_t = \sum_{c \in \mathcal{C}} \text{arr_flights}_{c,t}, \qquad \text{arr_del15}_t = \sum_{c \in \mathcal{C}} \text{arr_del15}_{c,t},$$

   where $\mathcal{C}$ denotes the set of reporting carriers at RDU in month t.

3. **Construct the target variable.** The monthly delay rate was computed as:

$$\text{delay_rate}_t = \frac{\text{arr_del15}_t}{\text{arr_flights}_t}.$$

   Expressing the outcome as a rate rather than a raw count controls for month-to-month variation in traffic volume and focuses the analysis on the incidence of delay conditional on operations.

4. **Create the time series object.** The resulting monthly series was converted to an R `ts` object with frequency 12 (monthly seasonality), starting in January 2008. The training set spans **January 2008–December 2018** (132 observations) and the test set spans **January–December 2019** (12 observations).

This aggregation strategy captures the overall delay experience at RDU but necessarily masks carrier-level heterogeneity, a limitation discussed in the final section of this report.

2.4 Processed Dataset

Table 2 displays the first ten observations of the processed monthly series used for modeling.

Table 2: First ten rows of the processed monthly RDU delay dataset.

Year	Month	Arriving Flights	Delayed Flights (15+ min)	Delay Rate
2008	1	5315	1122	0.2111
2008	2	4881	1279	0.2620
2008	3	5083	1313	0.2583
2008	4	4976	1152	0.2315
2008	5	5157	934	0.1811
2008	6	5261	1541	0.2929
2008	7	5533	1489	0.2691
2008	8	5317	1191	0.2240
2008	9	4717	762	0.1615
2008	10	4790	685	0.1430

2.5 Summary Statistics

Table 3 summarizes the central tendency and range of the main variables over the training period.

Table 3: Summary statistics for the monthly RDU delay series, January 2008-December 2018 (n = 132 months).

Variable	Min	Median	Mean	Max	SD
Arriving Flights (monthly)	2427.0000	3729.0000	3854.2000	6023.0000	877.3000
Delayed Flights 15+ min (monthly)	325.0000	728.5000	761.4000	1557.0000	276.5000
Delay Rate (monthly)	0.1045	0.1951	0.1949	0.3104	0.0429

Over the 132-month training window, RDU handled a stable but gradually growing volume of arriving flights per month, with approximately 18–20% of flights delayed by 15 minutes or more on average. The monthly delay rate exhibits moderate variation around this mean, consistent with a stationary series containing seasonal and irregular components. Detailed characterization of the temporal structure is presented in the following section.

3 Inspection & Analysis

3.1 ADF Test for Stationarity

The Augmented Dickey–Fuller test returns a test statistic of -4.2993 with a p-value of 0.01. Since the p-value is below 0.05, the null hypothesis of a unit root (non-stationarity) is rejected, indicating that the time series is likely stationary. This means the series does not exhibit a persistent stochastic trend and its statistical properties such as mean and variance remain relatively stable over time.

For forecasting, this suggests that non-seasonal differencing may not be necessary when fitting ARIMA-type models (d ~ 0). However, seasonal differencing may still be required if seasonal patterns are present, and models such as SARIMA, ETS, or other seasonal forecasting approaches can still be appropriate.

3.2 Autocorrelation

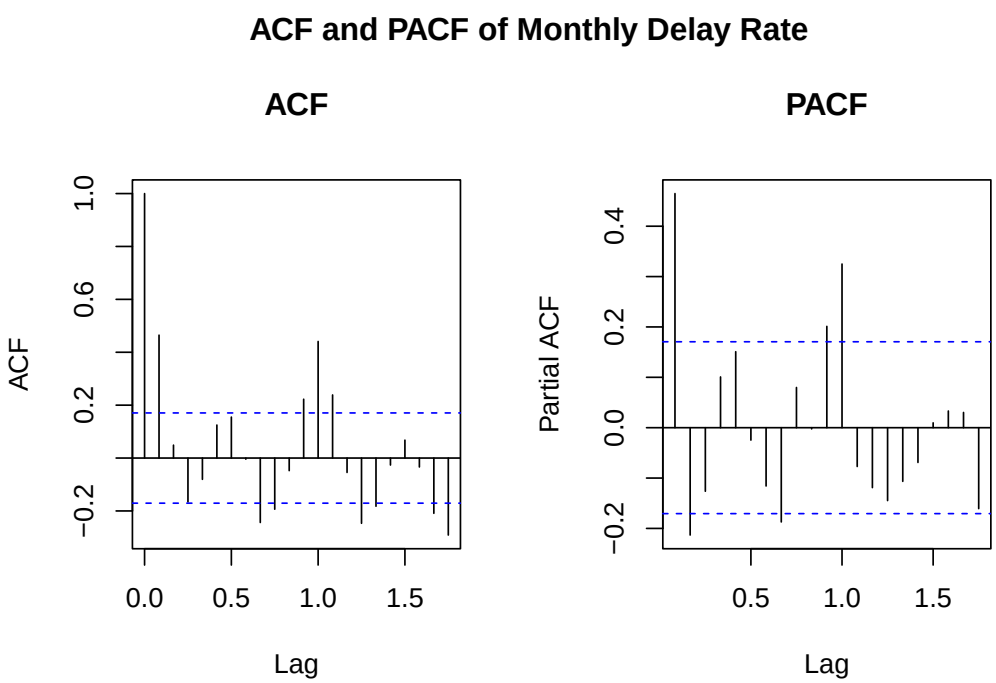

Figure 1: ACF and PACF of the monthly delay rate at RDU.

The ACF plot shows a strong positive autocorrelation at lag 1 and a noticeable spike around lag 12, suggesting both short-term dependence and yearly seasonality in the monthly series. The correlations generally decay rather than cutting off sharply, indicating the presence of autoregressive structure.

The PACF plot shows a significant spike at lag 1 and some additional seasonal spikes near lag 12, which further supports an autoregressive component and seasonal effects. Together, these patterns suggest the time series contains short-term autocorrelation and seasonal dependence, making seasonal ARIMA–type models appropriate for forecasting.

3.3 Decomposition

3.3.1 Data (Observed Series)

The observed data show monthly fluctuations in the air travel delay rate at RDU between roughly 0.10 and 0.30. The series displays noticeable short-term variability with recurring peaks and troughs across years,

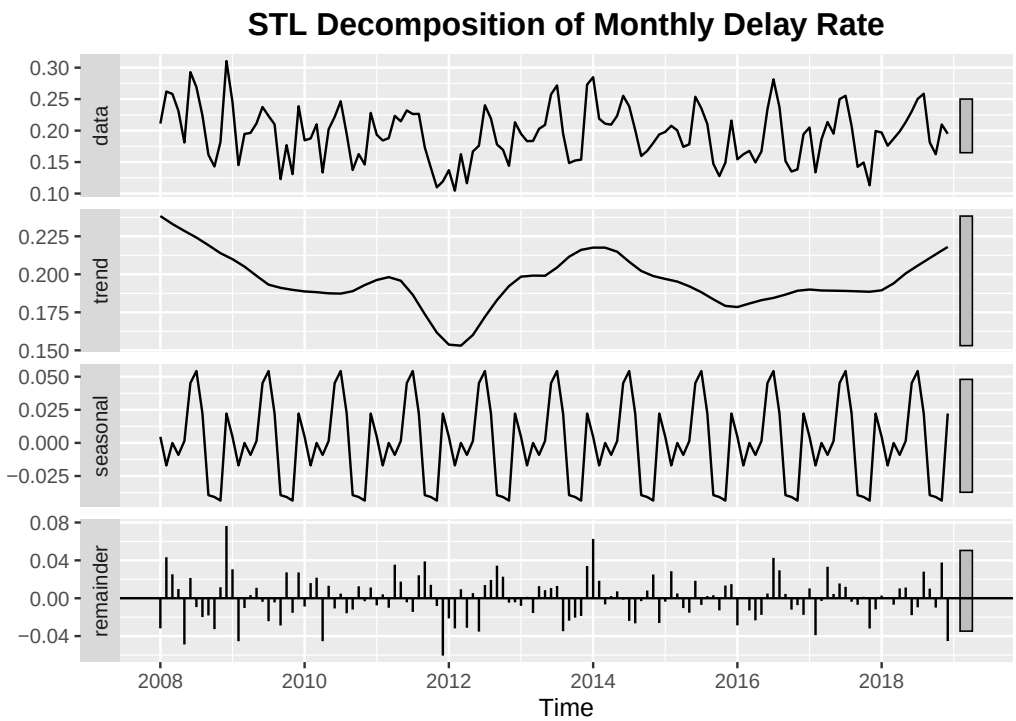

Figure 2: STL decomposition of the monthly delay rate at RDU into trend, seasonal, and remainder components.

suggesting the presence of seasonal patterns. Although the monthly values fluctuate considerably, there are also broader movements over time, indicating that both long-term trends and seasonal effects contribute to the overall behavior of the series.

3.3.2 Trend

The trend component reveals the longer-term evolution of delay rates. From 2008 to around 2010, there is a gradual decline in delay rates. Between 2011 and 2012, the trend drops more sharply and reaches its lowest point around 2012. After this, delays increase again from roughly 2012 to 2014, reflecting a recovery period. From 2015 to 2017, the trend shows a mild decline or plateau, indicating relatively stable conditions. Finally, from about 2018 onward, the trend rises again. This pattern reflects changes in demand and operational conditions at RDU over time. The initial decline likely corresponds to reduced air travel during the global recession, which decreased congestion and delays. As the economy recovered, passenger traffic increased between 2012 and 2014, placing more pressure on airport infrastructure and raising delay rates. Operational adjustments helped stabilize delays from 2015 to 2017, but rapid passenger growth by 2018 and 2019 appears to have again pushed delays upward.

3.3.3 Seasonal

The seasonal component shows a clear and repeating annual pattern. Delays tend to increase during the summer months and then decline sharply in the fall. The magnitude of the seasonal effect is relatively consistent across years, with an amplitude of roughly ± 0.04 to 0.05, meaning seasonality can change the delay rate by about 4–5 percentage points. This pattern is consistent with seasonal travel demand, where summer months often experience higher passenger volumes and potentially more weather-related disruptions. However, it is important to note that decomposition methods such as STL impose a seasonal structure by design, so some seasonal pattern will always appear. Therefore, the following will help validate the stable seasonal structure:

This result suggests that one seasonal difference (D = 1) is needed to remove seasonal dependence in the series, meaning observations are strongly related to values from the same month in the previous year.

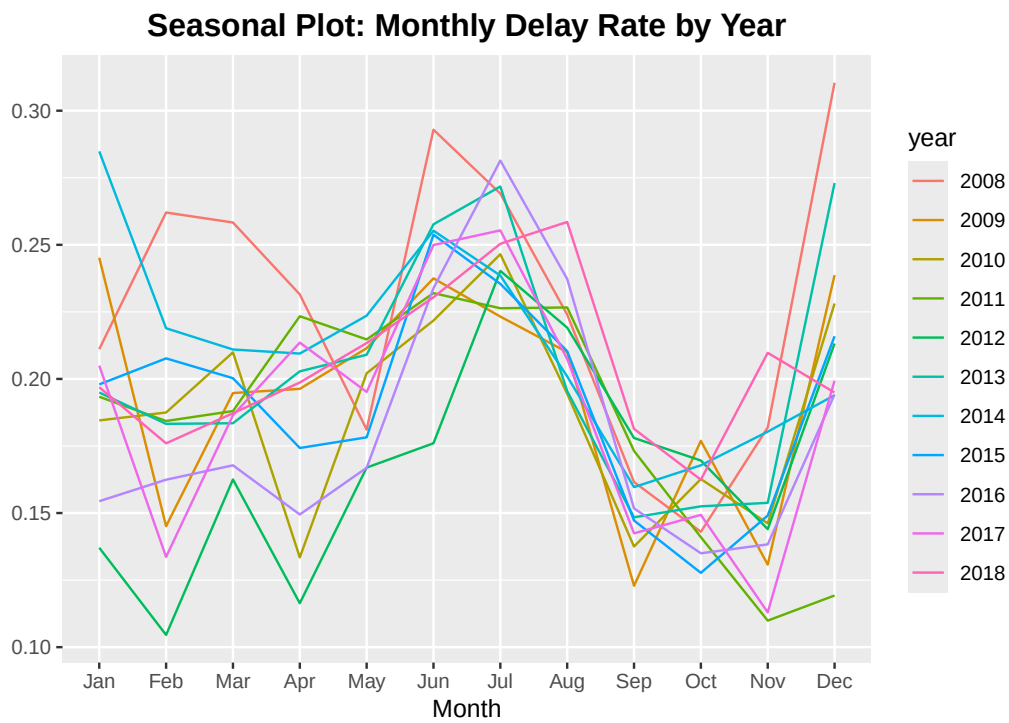

Figure 3: Seasonal plot of monthly delay rate by year, highlighting the consistent summer peak and fall trough across the training period.

The seasonal plot shows a consistent yearly pattern in the delay rate across different years. Most years follow a similar shape: delays tend to increase from late spring into the summer, peaking around June–July, and then decline sharply toward the fall months, with the lowest values typically occurring around September–November. Toward the end of the year, delays rise again slightly in December, which may reflect increased holiday travel and winter weather disruptions. Although the exact levels vary by year, the overall monthly pattern is fairly consistent, indicating a stable seasonal cycle in the data. Together, the seasonal plot and the seasonal differencing test confirm that the time series exhibits meaningful annual seasonality.

3.3.4 Remainder (Noise)

The remainder component represents the irregular variation not explained by the trend or seasonal components. In this decomposition, the residual values are mostly centered around zero and appear relatively small compared to the seasonal variation. Occasional spikes occur, indicating months with unusually high or low delay rates that cannot be explained by the general trend or seasonal cycle. Importantly, the remainder does not display obvious systematic patterns or persistent autocorrelation, suggesting that the trend and seasonal components capture most of the underlying structure of the series. This supports the interpretation that the identified seasonal pattern is likely meaningful rather than purely an artifact of the decomposition method.

4 Forecasting (Methods and Models)

4.1 Model 1: ETS

We fitted a total of 6 models to our data. The first model fitted was an ETS model, which explicitly accounts for the 3 main components: Error, Trend, Seasonality, assigning exponentially decreasing weights to older

data. The ETS model performs well with low bias and ~11% average percentage error, beats the naïve forecast clearly (MASE < 1, Theil's U < 1), and shows slightly better test accuracy than training, suggesting stable forecasting performance.

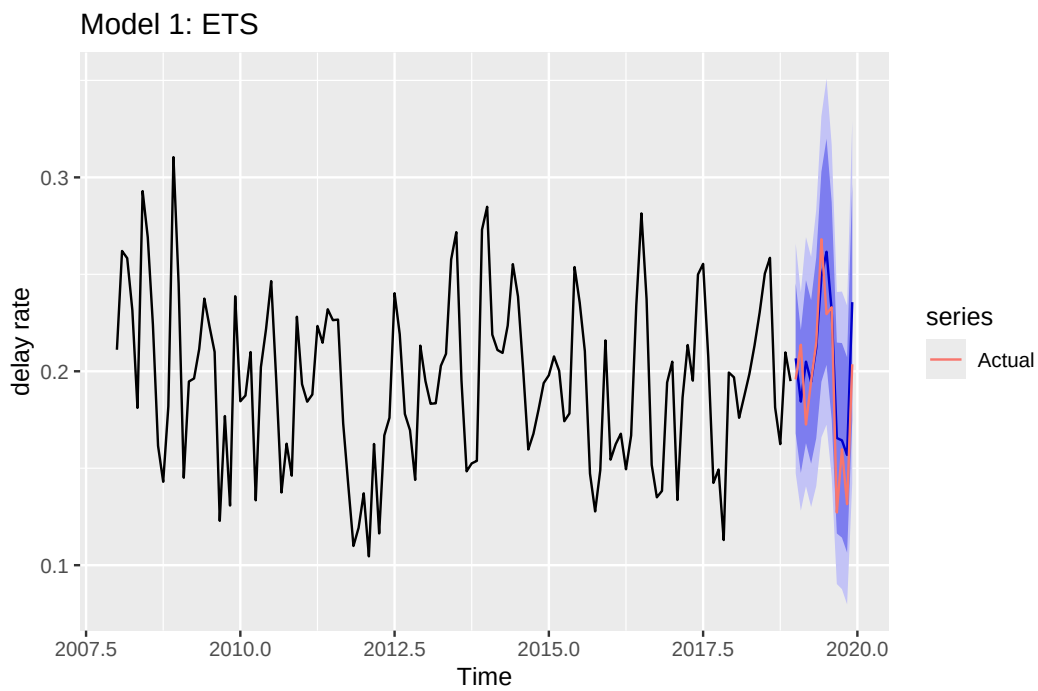

Figure 4: ETS forecast vs actual.

4.2 Model 2: SARIMA

We then fitted a SARIMA model with the parameters being determined using auto.arima. The resulting model has non-seasonal terms (1,0.0) and seasonal terms (2,1,0). Notably the P = 2 means there are 2 seasonal autoregressive terms, a given value depends on the values 12 and 24 months prior. The SARIMA model appears well calibrated with low bias, stable training vs test performance, and clearly better than a naïve benchmark, with average forecasting error around 11% of the delay rate.

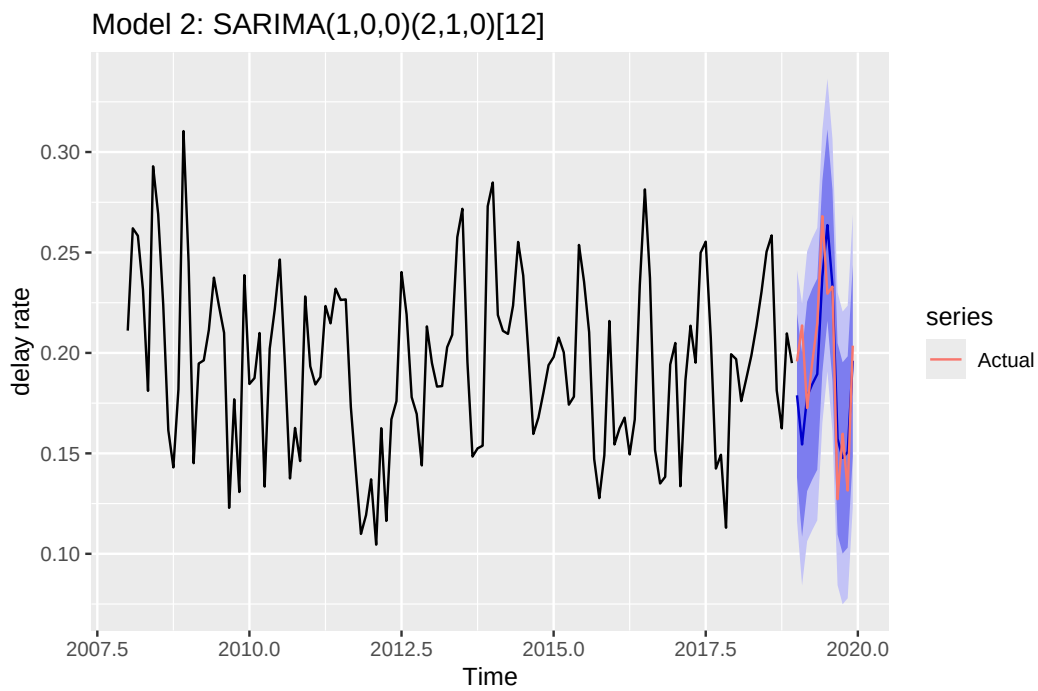

Figure 5: SARIMA forecast vs actual.

4.3 Model 3: ARIMA with Fourier Terms

We then employed more advanced models, first fitting an ARIMA model with Fourier terms. The resulting model uses Fourier terms as external regressors to capture the monthly seasonal pattern, replacing the need for a seasonal ARIMA structure. The remaining residuals are modeled as ARIMA(1,0,1).

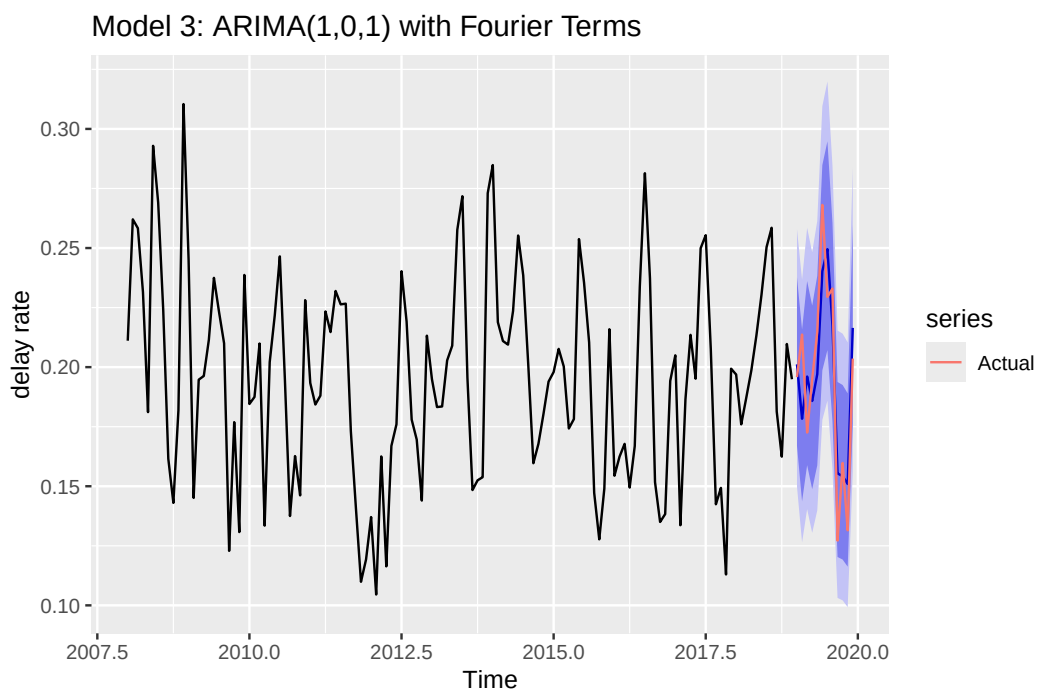

Figure 6: ARIMA with Fourier terms forecast vs actual.

4.4 Model 4: TBATS

In fitting a TBATS model to the data, a Box-Cox transformation with $\lambda = 0.621$ to stabilize variance was applied, no autoregressive structure was needed in this model, no trend was detected in the data, and monthly seasonality was modeled using 5 Fourier terms.

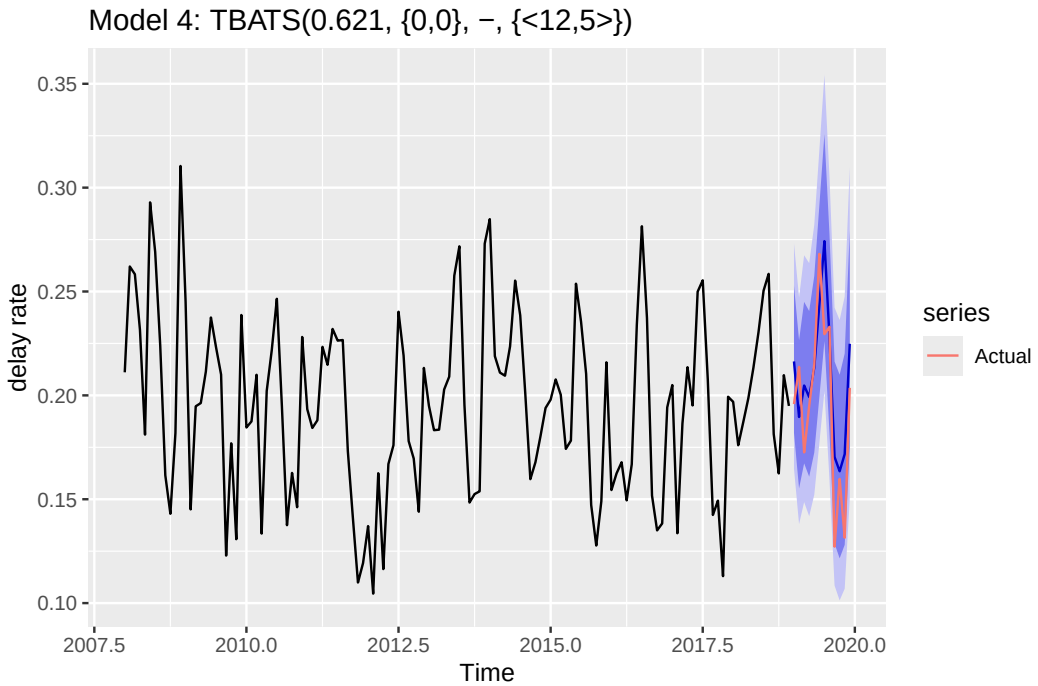

Figure 7: TBATS forecast vs actual.

4.5 Model 5: NNETAR

We then moved on to a feed-forward neural network autoregressive model with 1 non-seasonal lag term as the only autoregressive input and 6 neurons in a single hidden layer. It is worth noting that this model is the average of 20 networks, providing some more stability, although the model is still non-deterministic and so we set a random seed for reproducibility. The NNAR model has notably better training performance than testing performance, which means that there is risk of overfitting.

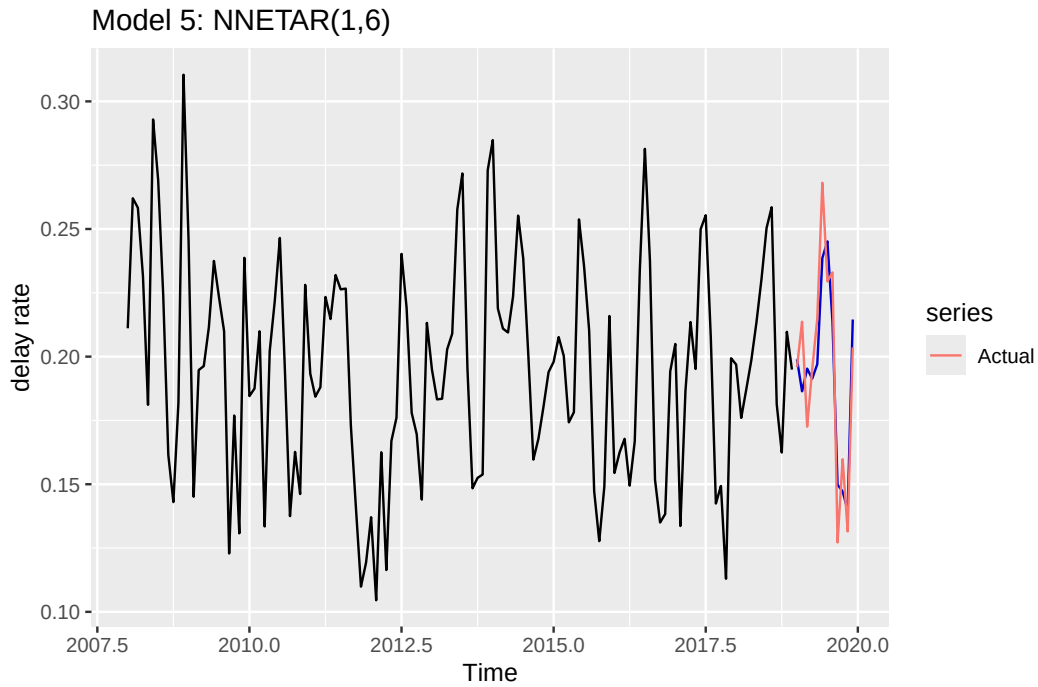

Figure 8: NNETAR forecast vs actual.

For all of the models so far, they fail to forecast the actual, initial, positive spike in the test data. Although this may not be due to flaws of our models, as instead they may represent real-life conditions that did not follow historical patterns. Also, our models tended to underestimate the amplitude of spikes in the actual test data. Excluding this, our models seem to be fairly close to the test data.

4.6 Model 6: Prophet

Lastly, we also applied Meta's Prophet model, which decomposes the time series into trend, seasonality (modeled by Fourier series), and holiday effects, fitting each component separately.

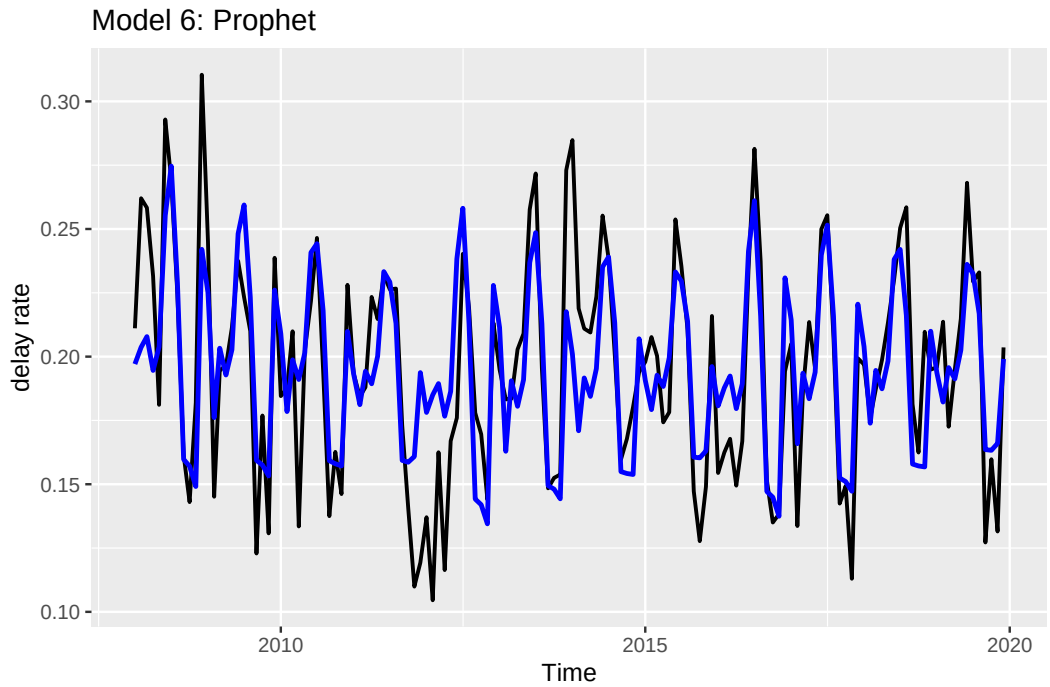

Figure 9: Prophet forecast vs actual.

Once again, the Prophet model tends to underestimate the amplitude of many of the spikes in the actual data, but with a few exceptions, its forecasts lie fairly close to the actual data.

4.7 Model Comparison

As shown in Figures 10 and 11, across our evaluation metrics, our best performing models seem to be the feed-forward neural network autoregressive model, the Prophet model, and the ARIMA model with Fourier terms. That being said, due to how close the performance of the models are, as well as the variability (due to not being deterministic) and the risk of overfitting of the neural network model, we might recommend using the Prophet model or the ARIMA model with Fourier terms.

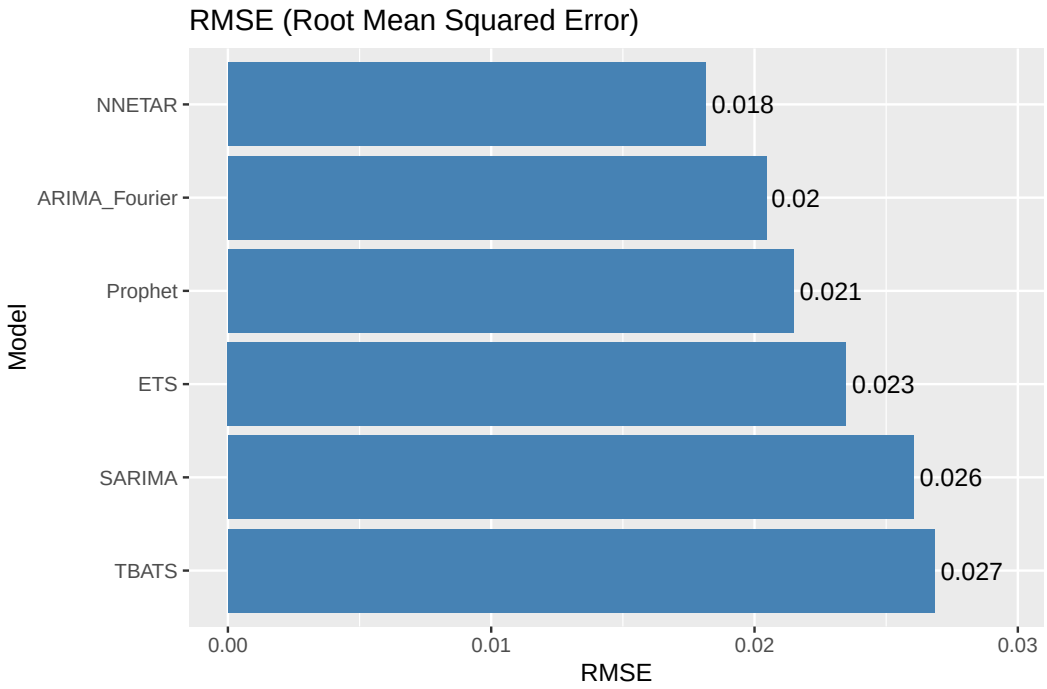

Figure 10: Comparison of the RMSE scores of each model.

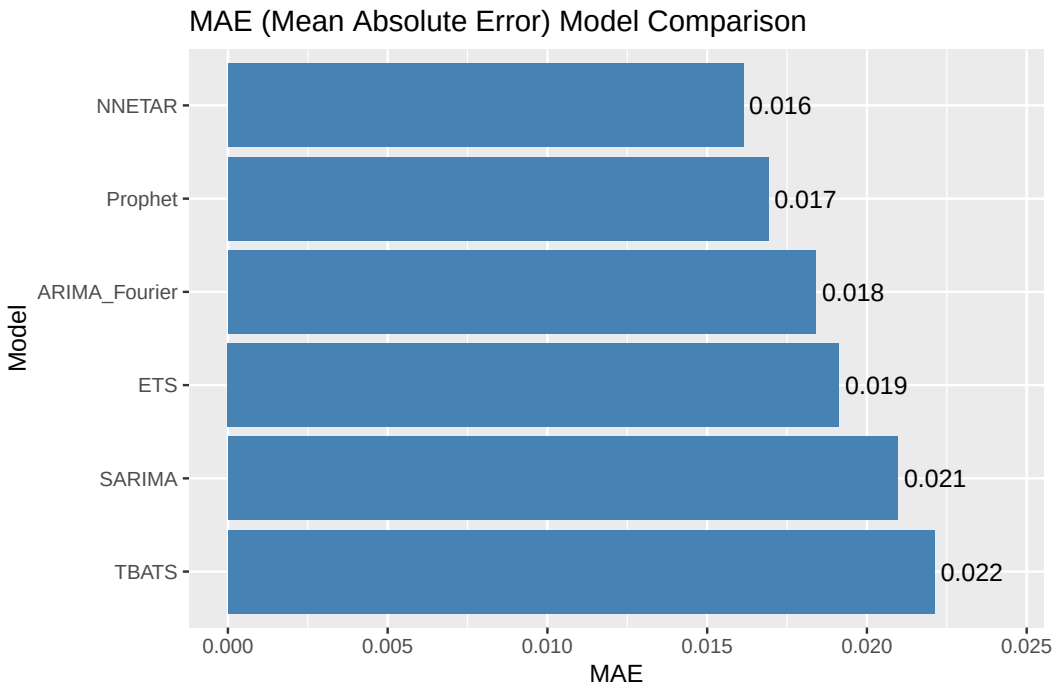

Figure 11: Comparison of the MAE scores of each model.

5 Summary and Conclusions

We built a monthly time series of arrival delay rates at RDU from 2008 to 2019 using BTS carrier-level data, trained six forecasting models on 2008–2018, and evaluated them on 2019. The series was stationary by ADF ($p = 0.01$), with clear yearly seasonality confirmed by both the STL decomposition and the seasonal plot. The remainder component fluctuated around zero without obvious leftover structure, which gave us

confidence the seasonality is a real signal rather than an artifact of the decomposition.

All six models landed in a fairly narrow performance range, with RMSE between 0.018 and 0.027 and MAE between 0.016 and 0.022. NNETAR scored the best, followed closely by Prophet and ARIMA with Fourier terms. Most models underestimated the amplitude of the spikes in 2019 and missed the early-2019 peak entirely, which likely reflects a mix of natural smoothing in time series models and one-off operational or weather events that don't show up in historical patterns.

For a final recommendation, we would favor Prophet or ARIMA with Fourier terms over NNETAR. Although NNETAR was slightly more accurate, it is non-deterministic and shows some signs of overfitting, which makes its performance harder to rely on in practice. Prophet and ARIMA with Fourier terms deliver comparable accuracy with stable, reproducible output, and both produce proper prediction intervals that are useful for operational planning. Between the two, ARIMA with Fourier terms offers a more transparent statistical structure, while Prophet handles trend changes and holiday effects more flexibly; either is a defensible choice for forecasting monthly delay rates at RDU.

Several limitations are worth noting. The monthly resolution hides within-month patterns like day-of-week or weather effects. The training window stops at 2018 to avoid COVID-19, but this means the models reflect a pre-pandemic regime that may no longer apply. Aggregating across carriers also masks carrier-level differences, since a few large airlines dominate the total signal. Future improvements could include adding exogenous variables (scheduled flights, weather, fuel prices), extending to post-COVID data once enough is available, or moving to a finer time resolution to capture operational detail that monthly data smooths over.

Statement on the Use of AI Tools

AI tools (OpenAI ChatGPT and Anthropic Claude) were used in a supporting role for this project. Specifically, AI assistance was used to check grammar and writing style, ensure the text adheres to a scientific rather than narrative tone, and help adjust the R Markdown and PDF formatting (YAML header, `kable` tables, LaTeX syntax).

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